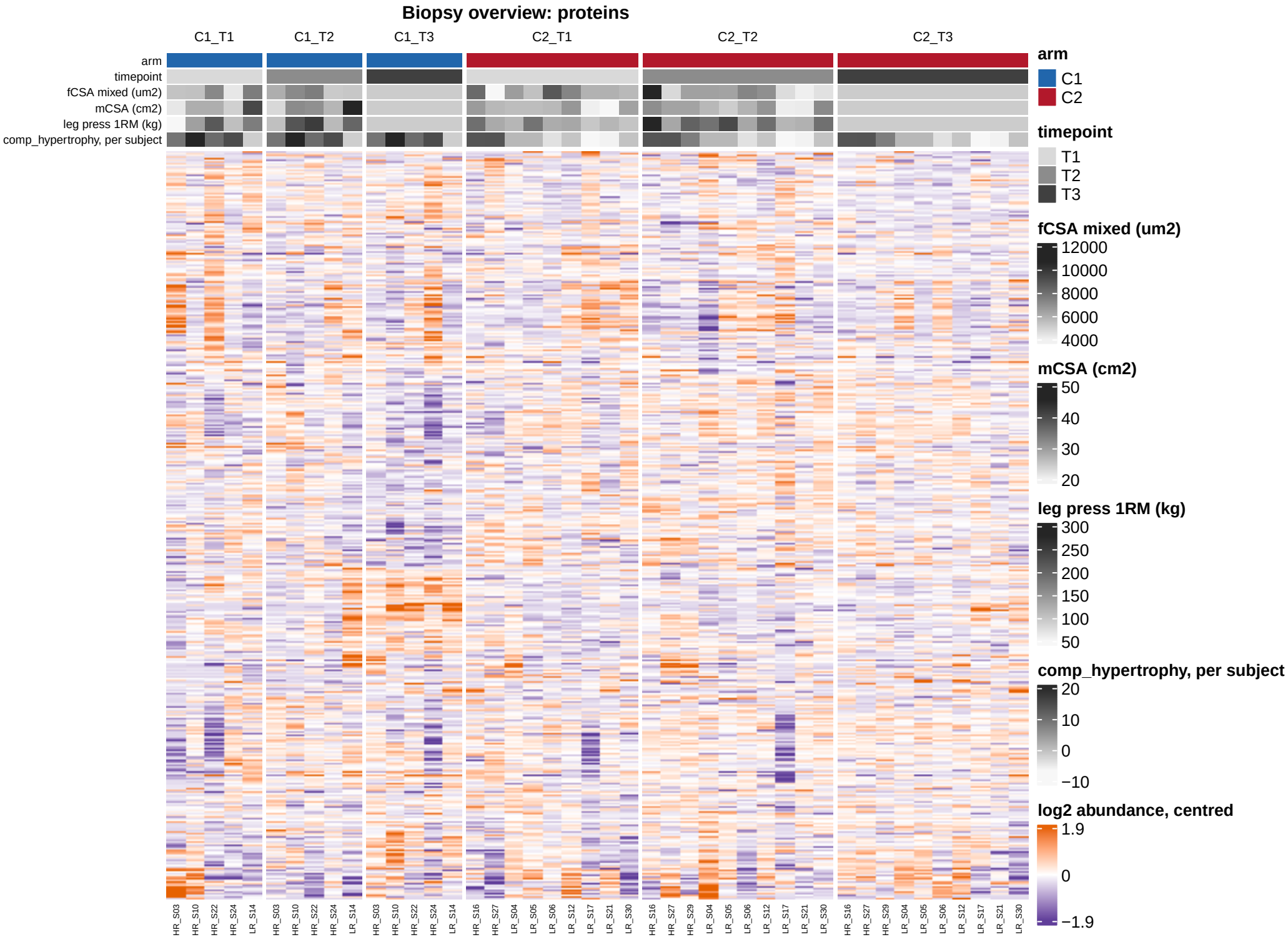
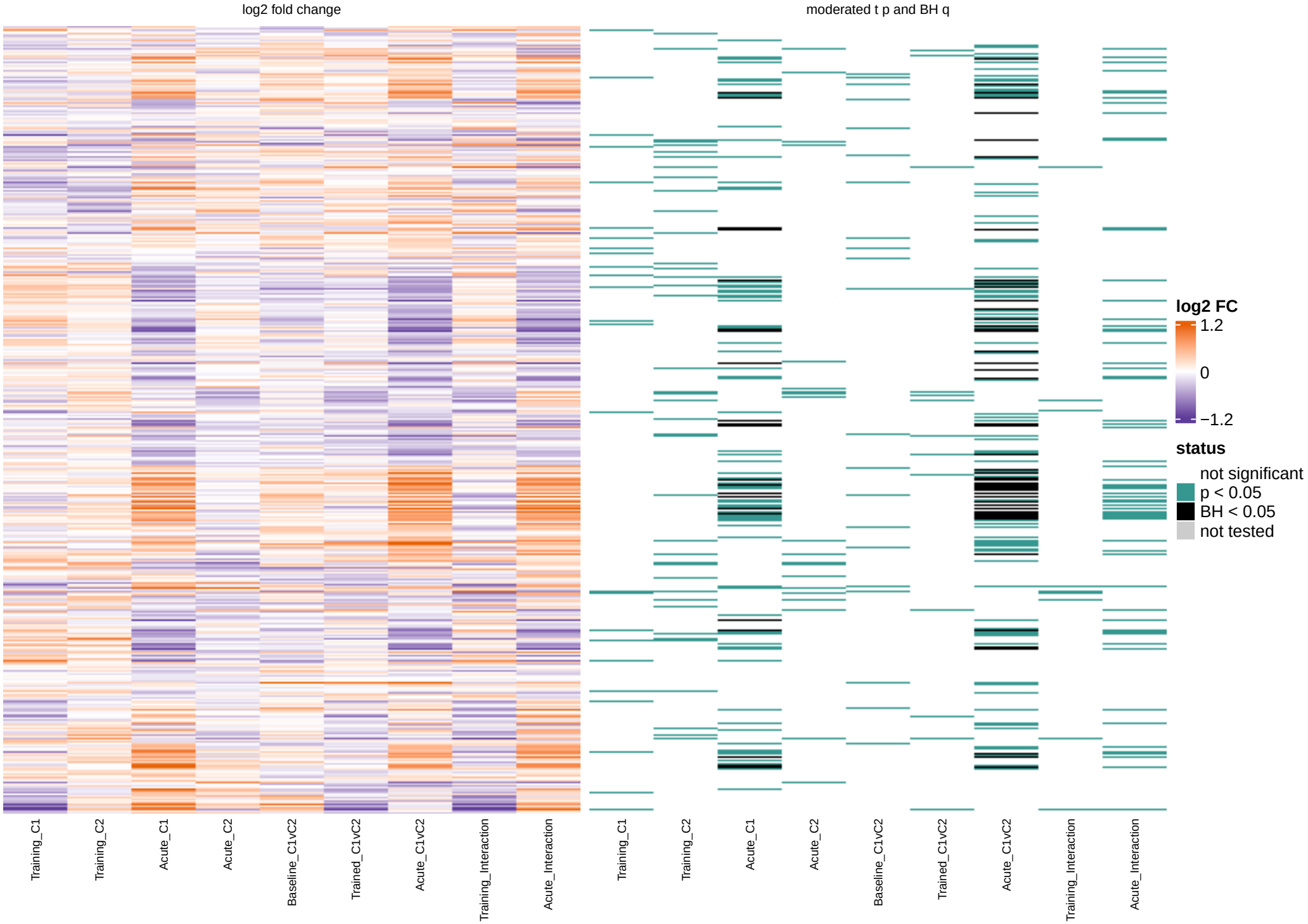



All 1,900 proteins by 44 biopsies, grouped by arm and time point; column labels are subjects. Fill: log2 abundance minus the protein's mean over biopsies. Colours stop at the 99th percentile of |value|, 1.9. Rows are ordered by average−linkage clustering (1 − Pearson r) of limpa's matrix, for display only. Grey in the trait tracks: not measured (T3 has no traits).

# Contrast overview: proteins



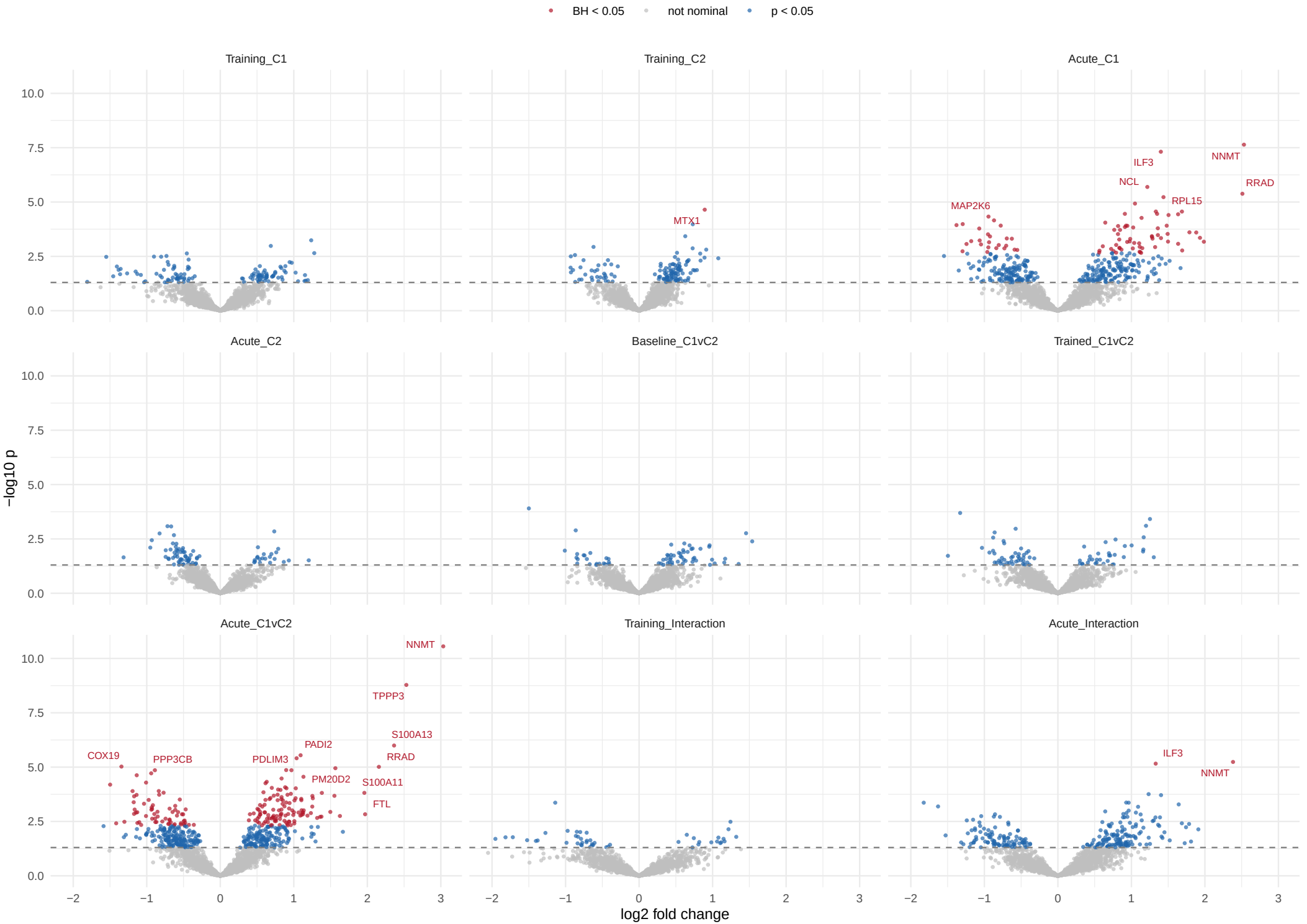

All 1,900 proteins by the nine contrasts. Fill: log2 fold change from the limma fit; status: p and BH q within contrast. Colours stop at the 99th percentile of |value|, 1.2. Rows are ordered by average−linkage clustering (1 − Pearson r) of limpa's matrix, for display only. Table: c_data/protein_contrasts.xlsx, protein_by_contrast.



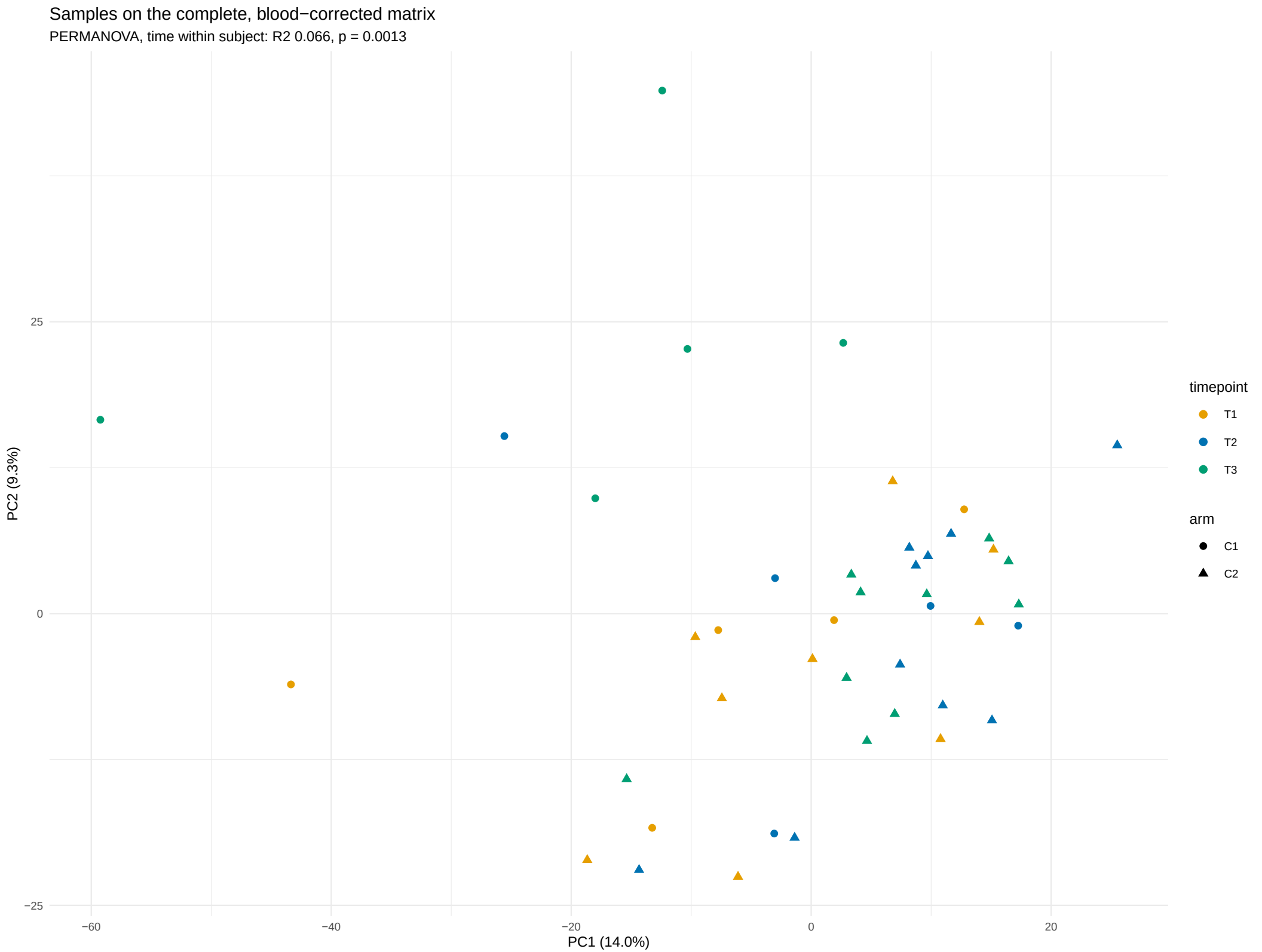

Dashed line: a flat null (tested / 20). A spike at zero on a flat background is signal; a slope up toward 1 is a conservative test. Table: c_data/protein_contrasts.xlsx, protein_by_contrast.

## Volcano per contrast

47 to 493 proteins at p < 0.05 per contrast; 271 at BH < 0.05 across all nine.




limma on limpa's quantification, subject blocked, blood as a covariate. Dashed line: p = 0.05. BH within contrast. Table: c_data/protein_contrasts.xlsx, protein_by_contrast.




Proteins scaled to unit variance. Table: c_data/protein_contrasts.xlsx, pca, permanova.